ActivIQ: Personal Fitness Chatbot

G1T3: Chang Jia Kai, Michael Suteja, Sim Qi Xun, Ng Hsien Ray, Chew Jun Liang Jerome

1. Problem Statement

Fitness information online is often generic, and requires searching across multiple sources before finding relevant advice, while general-purpose chatbots may be inaccurate, unsafe and require heavy computation. Personal trainers offer tailored guidance but may be unnecessary or inaccessible for users with simple fitness needs. Our project therefore proposes **ActivIQ**, a task-oriented conversational fitness assistant that uses Natural Language Processing (NLP) to understand users' fitness-related requests and return relevant, structured responses. It uses no external LLM APIs, keeping it reproducible and practical. We also test whether a small local LLM can replace manual data labelling.

2. Objectives

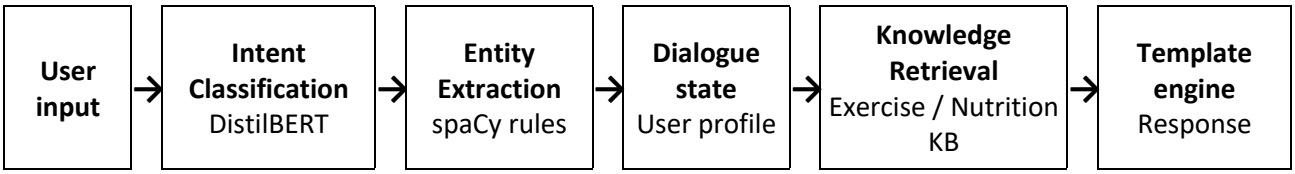
ActivIQ supports the six intents in our selected training dataset. Our objectives are to:

- Classify the six dataset intents with at least 90% accuracy and Macro-F1 ≥ 0.90
- Extract key entities (exercise, body part, equipment, goal, level, duration, food, nutrient) with F1 ≥ 0.85.
- Personalise responses using a lightweight profile maintained across conversation turns, targeting sub-second response time on a typical laptop CPU.
- Test whether a local LLM can label our data from 10 hand-labelled examples, and whether a small LoRA model trained on its labels can match DistilBERT and spaCy.

In scope: workout plans, exercise how-tos, basic nutrition information, feeling logs and light general chat.
Out of scope: medical diagnosis or treatment advice, open-ended generation, image or wearable input.

3. System Overview

ActivIQ uses a modular pipeline combining language understanding, dialogue management, information retrieval and controlled response generation[1].



DistilBERT was chosen because it keeps about 97% of BERT's performance while being 40% smaller and 60% faster[2]. spaCy rules suit a closed fitness vocabulary and avoid the need for costly annotation[3]. Retrieved exercise and nutrition information is then inserted into response templates, producing predictable and safe outputs, with no generative hallucination and no GPU needed at inference.

4. Implementation

4.1 Intent Classification

A DistilBERT model (distilbert-base-uncased) with a six-way classification head is fine-tuned using Hugging Face Transformers[4]. Input utterances will be tokenised to a maximum of 64 tokens, initial training settings use a learning rate of 2×10^{-5}, batch size 16 and 3 - 5 epochs, with values tuned on the validation set.

Intent label	Handles	Example utterance
generate_plan	Workout plans	"Design a 3-day full-body workout routine."
find_exercise	Exercise how-tos and routines	"How to do a Cossack squat."

[1] Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing* (3rd ed. draft). Chapters on dialogue systems and NLG

[2] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv:1910.01108.*

[3] Honnibal, M., Montani, I., Van Landeghem, S., & Boyd, A. (2020). spaCy: Industrial-strength Natural Language Processing in Python. https://spacy.io

[4] Wolf, T. et al. (2020). Transformers: State-of-the-Art Natural Language Processing. *Proc. EMNLP 2020: System Demonstrations.*

Intent label	Handles	Example utterance
get_nutrition_info	Nutrition facts and food questions	"How much vitamin K is in kale?"
log_feeling	How the user feels after training	"That was honestly the hardest workout I have ever done."
general_chat	Opinions, thanks and small talk	"Thanks for being so reliable."
out_of_scope	Unrelated topics	"How do I obtain an A+ for CS425?"

Softmax scores guide routing, with predictions below an initial 0.6 confidence threshold receiving a fallback response. A separate rule-based safety layer flags medical or high-risk queries and returns conservative responses instead of personalised advice.

4.2 Entity Extraction

spaCy's EntityRuler and PhraseMatcher run on lowercased, lemmatised text, using LOWER / LEMMA matching so plurals such as "squats" match their base forms. Phrase lists are seeded from the Free Exercise DB[5] and extended by hand. Matches are mapped to canonical values for knowledge base lookup.

Entity	Examples	Extraction method
Exercise	squat, bench press, plank	PhraseMatcher seeded from Free Exercise DB names
Body part	chest, quads, core	Phrase list with synonyms (e.g. "abs" → core)
Equipment	dumbbells, resistance bands	Phrase list
Goal	muscle gain, fat loss	Phrase list and token patterns
Level	beginner, intermediate	Phrase list
Duration	30 minutes, 4 weeks	Token pattern (number + time unit)
Nutrition	kale, vitamin K, protein	Phrase list from USDA food and nutrient names

4.3 Response Generation

Each intent has 3-5 template variants, chosen at random to avoid repetition. Slots such as {exercise}, {body_part} and {level} are filled from extracted entities and the user profile. Exercises are filtered by muscle, equipment and difficulty; nutrition queries look up the food in the USDA table. For workout plans, a rule-based planner distributes exercises across days, e.g. a 3-day full-body split for beginners.

If a required slot is missing, the bot asks a short follow-up such as "Which muscle group?". Out-of-scope queries use a fallback template, while medical queries return a fixed safety disclaimer.

4.4 Context Handling

The dialogue state is a Python dictionary, for example goal = muscle gain, level = beginner, equipment = dumbbells, last_intent = find_exercise. It is updated every turn: new entities overwrite stored values, follow-ups (e.g. 'What about chest?') inherit stored values, and users can reset their profile at any time.

4.5 Data

Dataset	Source	Size and preparation
Intent utterances	Public fitness-intent dataset[6]	1,200 utterances, six intents, 960/120/120 split.
Exercise knowledge	Free Exercise DB	800+ exercises with muscles, equipment, difficulty and instructions, cleaned into a JSON knowledge base.
Nutrition lookup	USDA FoodData Central SR Legacy[7]	Complete SR Legacy dataset: 7,793 foods with calories, macronutrients, vitamins, minerals and portions.

[5] Yuhonas. Free Exercise DB: Open Public Domain Exercise Dataset. GitHub. *github.com/yuhonas/free-exercise-db*

[6] Makwana, H. Fitness Intent Dataset. Hugging Face Datasets. *huggingface.co/datasets/harshmakwana/fitness-intent*

[7] U.S. Department of Agriculture, Agricultural Research Service. FoodData Central. *https://fdc.nal.usda.gov*

Dataset	Source	Size and preparation
Entity patterns	Hand-crafted	Phrase lists and regex; entities hand-labelled for 10 seed, 100 validation and some test utterances.

4.6 Automated Labelling and LoRA Comparison

The dataset has no entity labels, which would normally need manual annotation. We test whether a local LLM can do this: we hand-label 10 seed utterances and give them as examples to Qwen3.5-9B[8], run in 4-bit on Colab, which returns JSON labels.

The labelling prompt is chosen with Automatic Prompt Engineering (APE)[9]: the model writes candidate instructions from the seeds, each is scored by entity F1 on 100 hand-labelled validation utterances, and the best is compared with our own prompt. Intent agreement is also checked on the 960 training utterances.

The chosen prompt then labels the training set, and the smaller Qwen3.5-2B is fine-tuned on these labels with LoRA[10] to output intent and entities directly, and compared with DistilBERT + spaCy on the same test set for accuracy, F1 and response time. The main chatbot keeps DistilBERT + spaCy.

5. Evaluation Plan

Component	Metric	Target
Intent model	Accuracy and macro-F1 on held-out test set	≥ 90% / ≥ 90%
Entity extraction	Precision, recall, F1 on labelled test utterances	F1 ≥ 85%
End-to-end	Task success rate over multi-turn test scenarios	≥ 80%
Safety and speed	Unsafe/out-of-scope handling; response latency	100% handled; < 1 s
Auto-labelling	Entity F1 / intent agreement vs human labels	≥ 80% / ≥ 85%
LoRA model	F1 and latency vs DistilBERT + spaCy	Report comparison

6. Timeline and Responsibilities

Week	Phase	Key tasks	Lead
1	Data and intents	Finalise intents, build and clean dataset, fine-tune DistilBERT, set up Git.	Jia Kai, Hsien Ray
2	Understanding	Build entity rules, dialogue state and safety replies; APE search and LLM auto-labelling.	Michael, Hsien Ray
3	Generation	Build exercise database, templates, slot filling, full pipeline integration; LoRA fine-tuning.	Qi Xun, Jerome
4	Testing and delivery	Build Streamlit interface, run multi-turn tests, final metrics, report and demo.	Hsien Ray, Jerome

7. Risks and Limitations

- Rigid template replies; mitigated with several variants per intent.
- Small dataset and limited phrasing diversity; mitigated with paraphrase augmentation.
- Rule-based detection may miss unseen or implicit expressions; mitigated by iterative rule expansion.
- LLM labels may contain errors the LoRA model learns; we check them against human labels first.

8. Expected Outcomes and Future Work

Deliverables: a Streamlit prototype, trained intent and LoRA models, LLM-labelled entity data, knowledge base, templates and an evaluation report. **Future work:** OCR/VLM, wearable-device integration.

[8] Qwen Team (2026). Qwen3.5 small models (0.8B, 2B, 4B, 9B). Hugging Face Hub. huggingface.co/Qwen

[9] Zhou, Y. et al. (2023). Large Language Models Are Human-Level Prompt Engineers. ICLR 2023.

[10] Hu, E. J. et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.